

Epigenetic modifications as therapeutic targets

Theresa K Kelly^{1,2}, Daniel D De Carvalho^{1,2} & Peter A Jones¹

Epigenetic modifications work in concert with genetic mechanisms to regulate transcriptional activity in normal tissues and are often dysregulated in disease. Although they are somatically heritable, modifications of DNA and histones are also reversible, making them good targets for therapeutic intervention. Epigenetic changes often precede disease pathology, making them valuable diagnostic indicators for disease risk or prognostic indicators for disease progression. Several inhibitors of histone deacetylation or DNA methylation are approved for hematological malignancies by the US Food and Drug Administration and have been in clinical use for several years. More recently, histone methylation and microRNA expression have gained attention as potential therapeutic targets. The presence of multiple epigenetic aberrations within malignant tissue and the abilities of cells to develop resistance suggest that epigenetic therapies are most beneficial when combined with other anticancer strategies, such as signal transduction inhibitors or cytotoxic treatments. A key challenge for future epigenetic therapies will be to develop inhibitors with specificity to particular regions of chromosomes, thereby potentially reducing side effects.

Epigenetics encompasses the wide range of heritable changes in gene expression that do not result from an alteration in the DNA sequence itself. DNA methylation, the reversible post-translational modification of the range of histone variants, and nucleosome positioning collectively define the epigenetic landscape of a cell^{1,2}. DNA methylation occurs when a methyl group is added to the 5' position of the cytosine ring of CpG dinucleotides. Recently, methylation in embryonic stem cells was also suggested to occur at sites other than CpG dinucleotides, mainly on the cytosine of CHH or CHG trinucleotides (where H = A, C or T)³. In addition, it was recently shown that 5-methylcytosine can be converted into 5-hydroxymethylcytosine by members of the TET protein family⁴, mainly in embryonic stem cells and Purkinje cells⁵. The biological relevance of these recently described types of methylation is an area of active investigation. Histones can be covalently modified after translation by the addition of methyl, acetyl, phosphoryl, ubiquityl or sumoyl groups. Whether the modification facilitates or inhibits transcription depends on the histone residue modified and the type of modification. The localization of nucleosomes within genomic regulatory regions has an important role in creating environments that either permit or prevent transcription. Nucleosomes consist of DNA wrapped around a core of two copies of each of the H2A, H2B, H3 and H4 histone proteins, thus linking DNA methylation and histone modifications. The presence of particular variants of core histone proteins, such as H3.3 and H2A.Z, at specific genomic loci influences the stability of nucleosome occupancy. Thus, multiple levels of epigenetic control account for appropriate orchestration of gene expression in healthy cells and dysregulated gene expression in disease.

Here, we focus on recent examples in which epigenetic modifications have been used to evaluate disease risk, progression and clinical response. We aim to provide a broad overview of the accomplishments, remaining challenges and unrealized potential of epigenetic therapies in a range of diseases, with a particular emphasis on cancer.

Epigenetic disease mechanisms and their clinical relevance

Epigenetic aberrations have been well established in cancer^{6,7} and occur in several other diseases, including diabetes⁸, lupus⁹, asthma¹⁰ and a variety of neurological disorders^{7,11–13} (Table 1 and references within). In cancer cells, a global loss of DNA methylation (hypomethylation), particularly in gene bodies and intergenic regions (including repetitive elements) leads to genomic instability. This global hypomethylation is accompanied by increased *de novo* methylation (hypermethylation) of many promoters of tumor suppressors and other genes that are contained within CpG islands. This results in stable gene silencing (Fig. 1). In addition to changes in DNA methylation, cancer cells are characterized by a global loss of histone H4 Lys16 (H4K16) acetylation and H4K20 trimethylation. There is also increased expression of BMI1, a component of the polycomb repressive complex (PRC)-1, and EZH2, a histone-methylating component of PRC2, which both inhibit gene expression^{6,14}. Notably, recent evidence has shown that genes targeted by the PRC in embryonic stem cells are more likely than others to become methylated in cancer^{15–17}, suggesting that aberrant linkage between polycomb repression and the silencing of gene expression by DNA methylation may at least partly account for early changes seen during oncogenesis. Further understanding of the basis of this switch in epigenetic silencing mechanisms may provide new avenues to evaluate the tumorigenic potential of abnormal tissue.

Epigenetic modifications can be used to stratify disease subtypes, severity or treatment responsiveness¹⁸ and to predict clinical outcomes^{19,20}. H3 acetylation and H3K9 dimethylation can discriminate between cancerous and nonmalignant prostate tissue, and H3K4 trimethylation can predict the recurrence of prostate-specific antigen

¹Departments of Urology and Biochemistry and Molecular Biology, Norris Comprehensive Cancer Center, Keck School of Medicine, University of Southern California, Los Angeles, California, USA. ²These authors contributed equally to this work. Correspondence should be addressed to P.A.J. (pjones@med.usc.edu).

Published online 13 October 2010; doi:10.1038/nbt.1678

Table 1 Selected examples of epigenetic alterations associated with disease

Epigenetic aberration	Enzyme responsible	Disease	Epigenetic alteration	Comments	Reference
DNA methylation	DNMT1, DNMT3A, DNMT3B and DNMT3L	Rett syndrome	Inability to 'read' DNA methylation	<i>MECP2</i> mutation	11–13
		Diabetes	Hypermethylation of <i>PPARGC1A</i> promoter		8
		Cancer	Global hypomethylation, hypermethylation of some CpG island promoters, including CIMP		6,7,11
		Systemic lupus erythematosus	Hypomethylation of CpG islands at specific promoter regions	Decreased DNMT1 and DNMT3B expression	9
		ICF syndrome ATR-X syndrome	Hypomethylation at specific sites Hypomethylation of specific repeat and satellite sequences	<i>DNMT3B</i> mutation <i>ATRX</i> mutation	11–13 11,12
Histone acetylation	HATs and HDACs	Rubinstein-Taybi syndrome	Hypoacetylation	Mutation in gene encoding CBP, a known HAT	11–13
		Diabetes	Hyperacetylation at promoters of inflammatory genes		8
		Asthma	Hyperacetylation	Increased HAT activity and decreased HDAC activity	10
		Cancer	H4K16 acetylation loss	Hypomethylation of DNA repetitive sequences	6
Histone methylation	HMTs and HDMs	Cancer	H4K20me3 loss	Hypomethylation of DNA repetitive sequences	6
		Sotos syndrome	Decreased H4K20me3 and H3K36me3	Loss of function of NSD1, a HMT	113
		Huntington's disease	Increased H3K9me3 and possibly increased H3K27 trimethylation	Increased expression of the HMT ESET; enhanced PRC2 activity	12
miRNA expression	N/A	Cancer	Decreased miR-101	Increased EZH2, H3K27 trimethylation	74,87
			Decreased miR-143	Increased DNMT3A	88
			Decreased miR-29	Increased DNMT3A and DNMT3B	89
			Increased miR-21	Decreased PTEN	96
			Increased miR-155	Lower survival rates	95

ATR-X, alpha-thalassemia X-linked; CIMP, CpG island methylator phenotype; HAT, histone acetyltransferase; HDM, histone demethylase; HMT, histone methyltransferase; ICF, immunodeficiency, centromere instability and facial anomalies; me3, trimethylation.

accumulation after prostatectomy²¹. EZH2 expression is an independent prognostic marker that is correlated with the aggressiveness of prostate, breast and endometrial cancers²². Expression of the DNA repair gene O(6)-methylguanine-DNA methyltransferase (*MGMT*) antagonizes chemotherapy and radiation treatment²³. Accordingly, silencing of *MGMT* by endogenous hypermethylation is correlated with positive treatment response. Furthermore, epigenetic alterations can precede tumor formation and are thus potential diagnostic indicators of disease risk²⁴. For example, infection with *Helicobacter pylori* is associated with DNA hypermethylation of specific genes, which are often methylated in cancer²⁵. Thus, reversal of epigenetic alterations that occur as a result of an acute illness may prevent progression to a more chronic disease state.

The growing development of technologies to analyze the epigenome has led to the emergence of pharmacoeugenomics, the use of epigenetic profiles to identify molecular pathways most sensitive to cancer drugs²⁶ as a means of prioritizing therapeutic strategies. In non-small-cell lung cancer, an unmethylated *IGFBP3* promoter indicates responsiveness to cisplatin-based chemotherapy²⁷. A polymorphism in the gene encoding the CYP2C19*2 variant of a cytochrome P450 protein necessitates the use of higher doses of valproic acid (VPA) to achieve target plasma concentrations²⁸. Furthermore, epigenetic changes can be monitored to measure treatment efficacy and disease progression. Methylation of *PITX2* can be used to predict outcomes of individuals with early-stage breast cancer after adjuvant tamoxifen therapy²⁹. Patients with hypermethylation of the gene encoding p16 (*CDKN2A*) have lower recurrence rates

of bladder cancer compared to patients with no hypermethylation after interleukin-2 treatment³⁰. As epigenetic mechanisms determine which genes, and thus signaling pathways, can be activated, the presence of distinct modifications on specific genes and subsets of genes can aid at several steps in determining and monitoring optimal therapeutic approaches.

The reversibility of epigenetic modifications makes them more 'druggable' than attempts to target or correct defects in the gene sequence itself. Moreover, it is possible that cancer cells can become 'addicted' to the aberrant epigenetic landscape resulting from multiple epigenetic abnormalities³¹, rendering them more sensitive than normal cells to epigenetic therapy through a mechanism similar to an inverted oncogene addiction. A classic example of oncogene addiction is mesenchymal-epithelial transition factor (MET), a tyrosine kinase that acts as a receptor for hepatocyte growth factor and controls tissue homeostasis in normal cells³². MET can be aberrantly activated in cancer by ligand-dependent mechanisms or by overexpression³². Although MET has roles in both normal and cancer cells, the latter are more sensitive to MET inhibition owing to their greater reliance on MET signaling³². Thus, cancer cells become dependent (and consequently addicted) to increased activity of a few highly important oncogenes. It is possible that cancer cells undergo a parallel process by which they become dependent on aberrant silencing or inactivation of a few crucial tumor suppressor genes. As it is well known that several tumor suppressor genes are silenced in cancer by epigenetic mechanisms⁶, it is possible that cancer cells become addicted to their aberrant epigenetic landscape and consequently become more sensitive

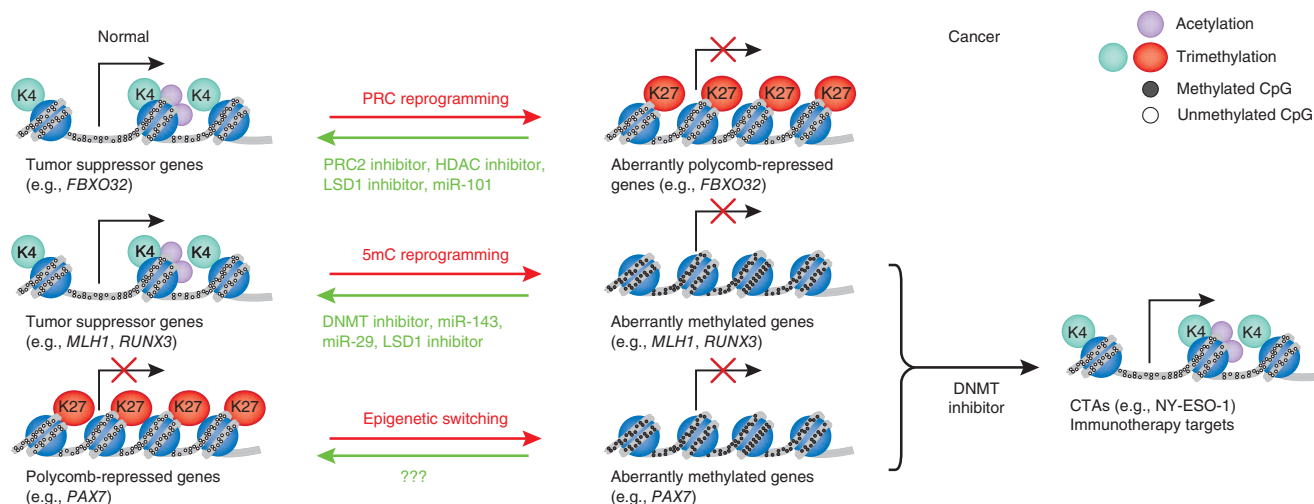


Figure 1 Epigenetic aberrations of CpG island promoters in cancer cells and the epigenetic therapies that target them. Tumor suppressor genes (such as *FBXO32*, *MLH1* and *RUNX3*) are expressed in normal cells and become silenced in cancer cells. This can occur either by PRC reprogramming (as for *FBXO32*), where the polycomb group protein EZH2 catalyzes the methylation of H3K27, or by 5-methylcytosine (5mC) reprogramming (as for *MLH1* and *RUNX3*) owing to *de novo* DNA methylation by DNMT3A and DNMT3B. Polycomb-mediated repression can be targeted by inhibitors of PRC2, such as DZNep, and re-expression of these genes can be enhanced by HDAC and LSD1 inhibitors allowing acetylation of H3 and H4 and methylation of H3K4, respectively. Polycomb-mediated repression can also be reversed by inducing miR-101 expression, which inhibits the expression and function of EZH2. 5mC reprogramming can be reversed, mainly by DNMT inhibitors, but also by re-expression of miR-143 and miR-29, two miRNAs that target *de novo* DNMTs. LSD1 inhibitors may also reactivate tumor suppressor genes by inhibiting DNMT1 stabilization, leading to loss of DNA methylation maintenance. Genes that are polycomb-repressed in normal cells (such as *PAX7*) can undergo epigenetic switching by DNA methylation, thus losing their plasticity during transformation. It is not known whether treatment of cancer cells with DNMT inhibitors alone can reverse epigenetic switching to restore the polycomb-repressed state or whether it will reactivate this set of genes. Cancer-testis antigens (CTAs, such as NY-ESO-1) can become silenced by DNA methylation in cancer. Treatment with DNMT inhibitors can induce CTA expression, allowing the immune system to recognize and kill the cancer cells. Red arrows represent epigenetic alterations during transformation; green arrows represent reversion of these alterations by epigenetic therapy.

to epigenetic therapy than normal cells. There is some evidence that cancer cells are preferentially affected by epigenetic therapies³³.

We next consider progress and remaining challenges in manipulating DNA methylation and histone modifications for therapeutic purposes, including microRNAs (miRNAs), which can also affect gene expression without altering DNA sequence and regulate as well as be regulated by epigenetic mechanisms. What are the merits and limitations of therapeutic strategies that intervene at these distinct levels of regulation of the epigenetic landscape? Moreover, how might they be used together or in combination with nonepigenetic therapies to prevent disease and remission?

DNA methylation

Cancer is characterized by global hypomethylation, with hypermethylation of a subset of gene promoters contained within CpG islands leading to gene silencing (Fig. 1)⁶. This hypermethylation has recently been described to extend past the boundaries of CpG islands into so-called DNA shores³⁴. DNA (cytosine-5)-methyltransferase (DNMT)-3A and DNMT3B are responsible for *de novo* DNA methylation patterns, which are then copied to daughter cells during S phase by DNMT1. DNA methylation inhibitors have been well characterized and tested in clinical trials³⁵. 5-Azacytidine (5-Aza-CR; Vidaza; azacitidine), a nucleoside analog that is incorporated into RNA and DNA, is approved to treat patients with high-risk myelodysplastic syndromes (MDS) and successful clinical results have recently been reported (Tables 2 and 3)³⁶. 5-Aza-2-deoxycytidine (5-Aza-CdR; Dacogen; decitabine) is the deoxy derivative of 5-Aza-CR and is incorporated only into DNA. At low doses, both azanucleosides act by sequestering DNMT enzymes after incorporation into DNA, leading to global demethylation as cells divide. At higher doses, they

induce cytotoxicity. Zebularine is a cytidine analog that acts similarly to 5-Aza-CR but has lower toxicity and greater stability and specificity³⁷. Another drug for which promising preclinical data are available is S110, a decitabine derivative with better stability and activity than 5-Aza-CdR (Fig. 2)³⁸. In addition to inhibiting DNMT activity, azanucleosides act through nonspecific mechanisms, which are likely to contribute to their clinical effectiveness.

Analysis of promoter DNA methylation can classify cancers^{26,39,40}, predict the progression of cancer^{41,42} and direct therapy^{43,44}. For example, DNA methylation of specific promoters may identify a subset of colorectal cancers that are responsive to 5-fluorouracil⁴³. Furthermore, use of DNA methylation inhibitors to reverse the silencing of *MLH1* restores sensitivity to cisplatin⁴⁵. This suggests that combining DNA methylation inhibitors with conventional chemotherapy drugs increases therapeutic efficacy. Successful conventional chemotherapy depends on activation of proapoptotic genes that respond to cytotoxic agents, leading to cell death. DNA methylation of these proapoptotic genes can prevent cell death, which in turn confers resistance to chemotherapy. Thus, reactivation of epigenetically silenced apoptotic genes should increase the efficacy of chemotherapy. For example, *APAF1* is silenced in metastatic melanoma cells, and treatment with 5-Aza-CdR restores expression and chemosensitivity⁴⁴. Conversely, methylation-induced silencing of DNA repair genes can be detrimental (by leading to microsatellite instability⁴⁶) or beneficial (by preventing the repair of genes targeted by chemotherapy, causing cells to undergo apoptosis rather than repair⁴⁷). Methylation-induced silencing of cancer-testis antigens, such as NY-ESO-1, can protect cancer cells from being recognized by T cells. Treating cancer cells with demethylating agents can induce the expression of these antigens, allowing recognition and killing by engineered cytotoxic

Table 2 Selected clinical trials of epigenetic cancer therapies with published findings

Epigenetic target	Agent	Phase of study	Disease	Findings	Number of subjects	Reference
DNMT inhibitor alone DNMTs	5-Aza-CR	2/3	MDS and AML	Complete remission in 10–17% and hematological improvement in 23–36%	309	114
		3	MDS	Better overall survival than with conventional care (24.5 vs. 15 months)	358	36
	5-Aza-CdR	2	MDS and CMML	Anti-MDS and anti-CMML activities with a safe toxicity profile; 34% of patients achieved complete response and 73% had objective response	95	115
HDAC inhibitor alone HDAC	Phenylbutyrate	1	MDS and AML	Well tolerated; no patients achieved complete or partial remission, although four achieved hematological improvement	27	116
	Vorinostat (SAHA)	1	Relapsed or refractory AML, CLL, MDS, ALL and CML	Seven of 31 AML patients showed hematological improvement, including two complete responses and two complete responses with incomplete blood count recovery	41	117
		1	Advanced solid and hematologic malignancies	One complete response (diffuse large B-cell lymphoma), three partial responses (cutaneous T-cell lymphoma)	73	118
Combination therapy DNMTs and HDAC	5-Aza-CR and VPA	1	Advanced solid cancers	Combination is safe; 25% of patients showed stable disease (median, 6 months)	55	110
	5-Aza-CR and phenylbutyrate	1	Refractory solid tumors	Combination is safe; no clinical benefit	27	112
	Vorinostat and doxorubicin	1	Solid tumors	Two of 24 showed partial responses (breast and prostate cancer) and two stable disease for more than 8 months (melanoma)	32	119
	Vorinostat plus carboplatin and paclitaxel	1	Advanced non-small-cell lung cancer	Better response ratio (34% versus 12.5%), progression-free survival (6 versus 4.1 months) and overall survival (13 versus 9.7 months) than with placebo plus carboplatin and paclitaxel	94	106

ALL, acute lymphoblastic leukemia; CLL, chronic lymphocytic leukemia; CML, chronic myelogenous leukemia; CMML, chronic myelomonocytic leukemia; SAHA, suberoylanilide hydroxamic acid.

T lymphocytes⁴⁸. This suggests the possibility of augmenting the efficacy of immunotherapy by combining it with drugs that modulate epigenetic regulation (Fig. 1).

Despite the clinical successes achieved with DNA methylation inhibitors, there is still considerable room for improvement. The available DNA methylation inhibitors block DNA methylation by trapping DNMT enzymes on DNA, preventing methylation at other genomic loci. Notwithstanding the therapeutic benefits of simultaneously counteracting the broad hypermethylation of tumor suppressor genes characteristic of most cancers, global hypomethylation may lead to activation of oncogenes and/or increased genomic instability. Moreover, DNA hypomethylation can activate promoters within repetitive elements. For example, hypomethylation of long interspersed nuclear element-1 can activate an alternative transcript of the MET oncogene in bladder cancer⁴⁹. Moreover, DNA methylation inhibitors have also been implicated in defects in memory-associated neural plasticity, suggesting a link between DNA methylation and neural plasticity associated with learning and memory⁵⁰.

Developing DNA methylation inhibitors that target specific genes or groups of genes would overcome these perceived risks of agents responsible for global DNA demethylation. Furthermore, because DNA methylation inhibitors act during the S phase of the cell cycle, they preferentially affect rapidly growing cells. This is advantageous when treating rapidly dividing cancer cells but may be less clinically

useful in treating diseases that are not characterized by rapid cell cycling. Moreover, the observation that levels of DNA methylation return to pretreatment levels upon withdrawal of azanucleoside¹¹ suggests a continual need for DNMT inhibition. Thus, despite the clinical success of DNA methylation inhibitors, their lack of specificity, cell cycle dependency and need for continuous administration leave room for the development of better therapies.

Histone modifications

Whereas DNA methylation is considered to be a very stable epigenetic modification, histone modifications are more labile. Levels of histone modifications are maintained by the balance between the activities of histone-modifying enzymes that add or remove specific modifications. As aberrant histone modification levels result from an imbalance in these modifying enzymes in diseased tissue, correcting the increased or decreased level of a particular enzyme should restore the natural equilibrium in the affected cells.

Cancer cells are characterized by dysregulation of histone methyltransferases and histone demethylases, overexpression of histone deacetylases (HDACs), and a global reduction in levels of histone acetylation^{6,14,51–53}. HDAC inhibitors have long been studied in the clinical setting as potential therapies (Fig. 2), and recent clinical trials of these agents have been extensively reviewed elsewhere (see also Tables 2 and 3)⁵⁴. HDAC inhibitors can also affect the acetylation

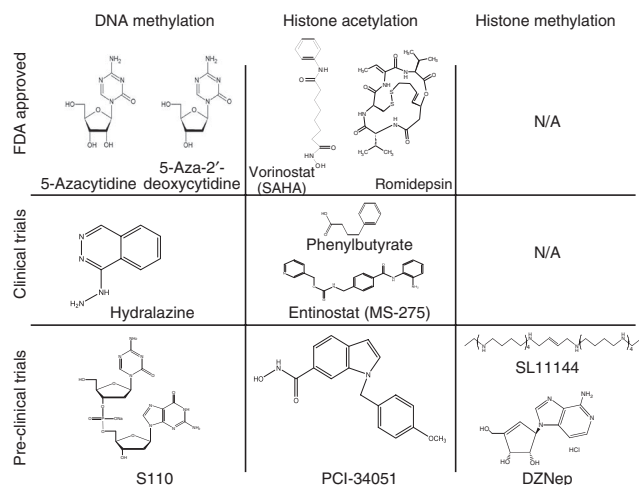


Figure 2 Chemical structures of selected compounds that target epigenetic modifications. Several molecules that target epigenetic alterations in pathological states are currently at different stages of drug development. The nucleoside analogs 5-azacytidine and 5-aza-2'-deoxycytidine are approved by the US Food and Drug Administration (FDA) to treat high-risk MDS, and successful clinical results have been reported. The drug hydralazine is currently being investigated in clinical trials as a putative demethylating agent against solid tumors. S110, a dinucleotide containing 5-aza-CdR, has been shown *in vitro* to demethylate DNA and is more stable than 5-aza-CdR because it is less sensitive to deamination by cytidine deaminase. Targeting of histone acetylation has also been a successful example of epigenetic therapy. Several HDAC inhibitors are FDA approved, including the hydroxamic acid-based compound SAHA and the depsipeptide romidepsin, whereas others are currently in clinical trials for cancer (phenylbutyrate and entinostat) and neurologic diseases (entinostat). New molecules targeting specific HDACs are under preclinical investigation (such as PCI-34051, which targets HDAC8). More recently, significant effort is under way to find new molecules able to target histone methylation. To our knowledge, no drugs targeting histone methylation are FDA approved or in clinical trials. Even so, preclinical trials suggest antitumor activity of the oligoamine analog SL11144, which inhibits LSD1, and the S-adenosylhomocysteine hydrolase inhibitor DZNep, which depletes cellular levels of PRC2 components.

of proteins other than histones, potentially leading to more global effects⁵⁴. Furthermore, because HDAC inhibitors only target ~10% of all acetylation sites⁵⁵, more work is necessary to understand the underlying basis for target specification of global and isoform-specific HDAC inhibitors. Substantial efforts are currently under way to find new molecules that can selectively inhibit specific HDACs^{56,57} and thus avoid the side effects that occur with a global HDAC inhibitor, including cardiac toxicity⁵⁴ and deficits in hematopoiesis⁵⁸ and memory formation^{59–61}. To date, specific inhibitors of HDAC6 (class II) and HDAC8 (class I) have been developed^{56,57}. When combined with a better understanding of the pathophysiology of diseases associated with alterations in HDACs, the development of specific HDAC inhibitors will allow more rational therapy and potentially reduce side effects. For example, the HDAC inhibitor PCI-34051, which is derived from a low-molecular-weight hydroxamic acid scaffold, selectively inhibits HDAC8 and induces apoptosis in T-cell lymphomas but not other tumor or normal cells. This indicates that HDAC8 has an important role in the pathophysiology of this disease and suggests that therapy with an HDAC8-specific inhibitor(s) can reduce undesirable side effects⁵⁷. Other HDAC inhibitors are selective to a group of HDAC isoforms, rather than a specific isoform, allowing their use for a wider range of diseases while minimizing side effects. For example, MGCD0103 (mocetinostat), which inhibits HDAC isoforms 1,

2 and 3 (class 1) and 11 (class 4), was shown in clinical trials to be tolerable and inhibit histone acetylation in patients with advanced solid tumors⁶². MGCD0103 was also shown to be safe and to have antileukemia effects⁶³. Although the identification of additional specific HDAC inhibitors will increase specificity and the possibility of personalized treatments, it may also limit the likelihood of their successful incorporation into combinatorial therapies.

Histone methyltransferase and demethylase enzymes are generally more specific than HDACs in that they target fewer residues⁶⁴. However, like HDACs, lysine and arginine methyltransferase enzymes also methylate proteins other than histones^{65,66}. A great deal of effort is under way to find drugs able to revert specific histone methylation marks or to selectively target histone methyltransferases or histone demethylases. In this regard, a new class of oligoamine analogs was recently found that act as potent inhibitors of lysine-specific demethylase-1 (LSD1; **Fig. 2**). LSD1 targets the activating H3K4 mono- and dimethylation mark but can also target the repressive H3K9 dimethylation (H3K9me2) mark when complexed with the androgen receptor^{51,67}. Treatment of colon cancer cells with LSD1 inhibitors (such as SL11144) increases H3K4 methylation, decreases H3K9me2, and restores expression of *SFRP2* (ref. 68), indicating context specificity of LSD1 and its inhibitors. LSD1 inhibition in neuroblastoma results in decreased proliferation *in vitro* and reduced xenograft growth⁶⁹. Notably, LSD1 can also demethylate DNMT1, resulting in destabilization and loss of global maintenance of DNA methylation⁷⁰. The ability of LSD1 to affect both histone and DNA methylation makes it a promising target for epigenetic therapy.

The repression mediated by the H3K27 trimethylation (H3K27me3) mark occurs through the actions of two multisubunit complexes, PRC1 and PRC2. The H3K27me3 mark deposited by EZH2 is recognized and bound by PRC1, which can further recruit additional proteins to establish a repressed chromatin configuration⁶. Gene promoters that are marked by PRC2 (that is, polycomb target genes) in embryonic stem cells have recently been shown to be far more likely than other genes to become methylated in cancer^{15–17}. Similarly, polycomb targets in normal prostate cells also become methylated in prostate cancer⁷¹. Thus, alterations in chromatin structure do not always coincide with changes in gene expression associated with disease. Instead, DNA methylation replacement of polycomb repressive marks 'locks in' an inactive chromatin state through a process called epigenetic switching⁷¹. Although the mechanism underlying the predisposition of polycomb targets for DNA methylation is not fully understood, some links have recently been uncovered. CBX7, a component of the PRC1 complex, can directly interact with DNMT1 and DNMT3B at polycomb target genes⁷².

Although drugs that target histone methylases and demethylases have considerable potential, more work is necessary to determine their specificities and the stabilities of the changes they effect. There are currently no such drugs in clinical trials. Preclinical studies suggest that the S-adenosylhomocysteine hydrolase inhibitor 3-deazaneplanocin A (DZNep) shows the most promise (**Fig. 2**). DZNep depletes cellular levels of PRC2 components (EZH2, EED and SUZ12) and consequently reduces H3K27me3 levels and induces apoptosis in breast cancer, but not normal, cells⁷³. The effect of DZNep is similar to that observed when EZH2 is depleted by RNA interference, suggesting that this drug is more effective in cancers of the prostate and breast, which rely on abnormally high EZH2 expression levels⁷⁴. In contrast, a subsequent study showed that DZNep also decreases H4K20me3. This demonstration that DZNep lacks specificity and acts more as a global histone methylation inhibitor underscores the need for further development of histone methylation inhibitors⁷⁵.

Table 3 Epigenetic cancer therapies under commercial development (either in safety and efficacy trials or approved)

Drug	Sponsor	Indication	Clinical status
DNMT inhibitors			
5-Aza-CdR (Dacogen)	Eisai (Tokyo)	MDS AML 1 st line CML	Approved May 2006 Phase 3 in 480 patients Phase 2 in 19 patients
5-Aza-CR (Vidaza)	Celgene (Summit, NJ, USA)	MDS AML Hematologic cancer	Approved May 2004 Phase 3 targeting 480 patients Phase 2
S110 (dinucleotide prodrug of decitabine)	SuperGen (Dublin, CA, USA)	MDS and AML	New Drug Application
HDAC inhibitors			
Romidepsin (Istodax; a cyclic depsipeptide)	Celgene	CTCL NHL	Approved November 2009 Phase 2
Vorinostat (Zolinza; suberoylanilide hydroxamic acid)	Merck (Whitehouse Station, NJ, USA)	CTCL	Approved October 2006
Vorinostat + bortezomib (Velcade)		Mesothelioma MDS, NHL, brain cancer and NSCLC	Phase 3 targeting 660 patients Phase 2
Panobinostat (LBH589; hydroxamate analog)	Novartis (Basel)	Multiple myeloma Hodgkin's lymphoma CML, AML and MDS	Phase 2 and 3 targeting 742 patients Phase 3 in 367 patients Phase 2/3
Panobinostat + bortezomib + dexamethasone		Multiple myeloma	Phase 3 targeting 676 patients
Belinostat (PXD10; hydroxamate analog)	Spectrum Pharmaceuticals (Irvine, CA, USA)	AML, CTCL, MDS, NHL and ovarian cancer	Phase 2
Mocetinostat dihydrobromide (MGCD0103; aminopyrimidine analog)	MethylGene (Montreal, QC, Canada)	AML, CLL, Hodgkin's lymphoma, NHL, pancreatic cancer and thymic carcinoma	Phase 2
Entinostat (SNDX-275; synthetic benzamide derivative)	Syndax Pharmaceuticals (Waltham, MA, USA)	Breast cancer, Hodgkin's lymphoma and NSCLC	Phase 2
PCI-24781 (CRA-024781; hydroxamic acid derivative)	Pharmacyclics (Sunnyvale, CA, USA)	Hematologic cancer and sarcoma	Phase 1/2
Other			
¹³¹ I-conjugated monoclonal antibody targeting DNA-histone H1 complexes (Cotara)	Peregrine Pharmaceuticals (Tustin, CA, USA)	Glioblastoma multiforme	Phase 2 in 40 patients

CLL, chronic lymphocytic leukemia; CML, chronic myelogenous leukemia; CTCL, cutaneous T-cell lymphoma; NHL, non-Hodgkin's lymphoma; NSCLC, non-small-cell lung cancer. Sources: BioMedTracker, Thomsen Pharma Partnering and PubMed.

EZH2 activity can also be regulated by signaling cascades. For example, AKT phosphorylates EZH2 at Ser21, suppressing its methyltransferase activity and thereby reducing levels of H3K27me3 (ref. 76). The frequency of H3K27 trimethylation can be restored using LY294002, an inhibitor of the phosphatidylinositol-3-kinase and AKT pathway, opening a new therapeutic opportunity to repair epigenetic alterations by targeting upstream signaling pathways. Furthermore, in prostate cancer, the oncogenic ETS transcription factor ERG can bind to the *EZH2* promoter and induce overexpression. Thus, pharmacological disruption of ERG activity could reduce the EZH2 overexpression observed in cancer⁷⁷. EZH2 is a particularly important example because it is frequently overexpressed and aberrantly targeted to genes in cancer⁷¹, a process termed PRC reprogramming (Fig. 1).

G9a and G9a-like protein (GLP) are histone methyltransferases that catalyze H3K9 dimethylation and are often overexpressed in tumors⁷⁸. Knockdown of G9a in prostate cancer cells indicates a crucial role for this protein in regulating centrosome duplication and chromatin structure. The likely importance of G9a in perpetuating the malignant phenotype and its promise as a target in cancer therapy⁷⁹ have generated substantial interest in developing G9a and GLP inhibitors. Thus far, the most efficient inhibitor is BIX-01294, a diazepine-quinazoline-amine derivative that transiently reduces global H3K9me2 levels in several cell lines⁸⁰. BIX-01294 binds to the SET domain of GLP in the same groove at which the target lysine (H3K9) binds. This prevents the binding of the peptide substrate and, consequently, the deposition of methylation marks at H3K9 (ref. 81).

Several other histone methyltransferases and demethylases have also been associated with diseases, making them potential targets for epigenetic therapy. For instance, MMSET, a H4K20 methyltransferase, is overexpressed in myeloma cell lines and is required for cell viability⁸².

SMYD3, a H3K4 methyltransferase, is also highly expressed in cancer and seems to have a role in carcinogenesis as a coactivator of estrogen receptor- α ⁸³. Expression of GASC1, an H3K9 and H3K36 demethylase, is often amplified in cancer, and its inhibition decreases rates of cell proliferation⁸⁴.

Although the challenges associated with targeting specific histone modifications have not prevented considerable clinical success with this group of targets, it seems likely that therapeutics capable of targeting specific histone-modifying enzymes could retain or increase therapeutic success rates while decreasing side effects resulting from the lack of specificity. In contrast, targeting individual histone modifying enzymes may decrease clinical efficacy if histone-modifying enzymes not targeted by the drug in question compensate for any changes and thereby confer drug resistance. Designing personalized cocktails of inhibitors based on an individual's need may help overcome the potential problems of compensation and resistance.

MicroRNAs

Small, noncoding miRNAs are able to induce heritable changes in gene expression without altering DNA sequence and thus contribute to the epigenetic landscape. In addition, miRNAs can both regulate and be regulated by other epigenetic mechanisms. Expression of miRNAs is dysregulated in several diseases, including cancer⁸⁵ and certain neurodegenerative disorders⁸⁶. For example, miR-101 targets EZH2 for degradation and is downregulated in several types of cancer, leading to increased EZH2 expression (and consequently higher H3K27me3 levels) and decreased expression of tumor suppressor genes^{74,87}. Restoring expression of miR-101 leads to reduced H3K27me3 and inhibits colony formation and cancer cell proliferation^{74,87}. Expression of miR-143 in colorectal cancer cells⁸⁸ and the miR-29 family in lung

cancer cells⁸⁹ reduces DNMT3A and DNMT3B levels, respectively, and results in decreased cell growth and colony formation. Treatment of cells with 5-Aza-CdR and 4-phenylbutyric acid results in miR-127 activation, which in turn downregulates the *BCL6* oncogene in bladder cancer cells⁹⁰. In fact, treatment with 5-Aza-CdR alone is sufficient to reactivate miR-148a, miR-34b/c and miR-9—a group of miRNAs capable of suppressing metastasis⁹¹. In addition to inducing aberrantly repressed miRNAs using epigenetic drugs, replacement gene therapy may also be useful in reestablishing miRNA expression. Viral vectors generated by cloning individual or groups of human miRNAs have been successful in preclinical assays using a mouse model of hepatocellular carcinoma, in which miR-26a expression from an adeno-associated virus results in apoptosis and inhibition of cancer cell proliferation in the absence of toxicity⁹². Gene therapy using miRNAs has an advantage over conventional RNA interference in that it is unlikely to generate a strong type I interferon response because double-stranded RNA is not introduced to the cell⁹³.

Abnormally high expression of miRNAs can be targeted using recently developed locked nucleic acid (LNA)-modified phosphorothioate oligonucleotide technology. LNA-modified oligonucleotides contain an extra bridge in their chemical composition, leading to enhanced stability compared to their unmodified counterparts. These LNA-modified phosphorothioate oligonucleotides can generate miRNAs, creating LNA-anti-miRNAs that can be delivered systemically. In preclinical assays with primates, intravenous injections of LNA-anti-miRNA complementary to the 5' end of miR-122 antagonized liver-specific expression of this miRNA without toxicity⁹⁴. Phase 1 trials based on these promising results are currently under way. LNA-anti-miRNAs may be used to target aberrantly expressed miRNAs in other diseases, such as cancer. For example, miR-155 is upregulated in lung adenocarcinoma compared to noncancerous lung tissue, and patients with higher miR-155 expression have lower survival rates than do patients with lower miR-155 expression. This suggests that miR-155 is a promising target for LNA-anti-miRNA therapy (Table 1)⁹⁵. Several other miRNAs are upregulated in cancer and could theoretically be used as LNA-anti-miRNA targets. For example, miR-21 is upregulated in several types of cancer (lung, breast, colon, gastric and prostate carcinomas; endocrine pancreatic tumors; glioblastomas; and cholangiocarcinomas) and targets the tumor suppressor PTEN (Table 1)⁹⁶. Thus, miRNAs can both alter the epigenetic machinery and be regulated by epigenetic alterations. This creates a highly controlled feedback mechanism, making it a suitable target for epigenetic therapy and possibly an epigenetic drug itself.

One unique advantage of targeting miRNAs is the ability of one miRNA to regulate several target genes and multiple cellular processes. In that way, if the level of one or a few miRNAs has changed in a pathological state, several different pathways could consequently be altered. Rather than trying to identify and directly target the proteins in multiple pathways, it would be more effective to restore the physiological level and functions of the dysregulated miRNA(s). This clinical potential highlights the importance of better understanding miRNA profiles in healthy and diseased tissues in order to develop better therapeutic strategies. Furthermore, multiple miRNAs that target different steps of an overactive pathway could be combined to increase efficacy and allow for customization of therapies to individual patients. Although the unique composition of miRNA-based therapy provides many benefits, additional research is necessary to determine the best method of delivery and increase miRNA stability to ensure efficacy.

Combined epigenetic therapies

The presence of multiple epigenetic aberrations in a single tissue, the ability of diseased cells to develop resistance, and the discovery that

common sets of genes are regulated by distinct epigenetic mechanisms at different biological stages collectively point to the likely feasibility of combinatorial approaches to target epigenetic modulators. Efforts for more than 25 years to enhance therapeutic efficacy by combining epigenetic strategies^{97,98} have revealed both additive and synergistic effects, depending on the targets^{11,52}. Extensive work on the clinical benefits of combined DNMT and HDAC inhibition has been comprehensively reviewed elsewhere^{6,46,54,99,100}. A recent phase 2 multicenter study examining the combination of 5-Aza-CR and the HDAC inhibitor VPA in patients with higher-risk MDS found that therapeutic levels of VPA may increase the efficacy of 5-Aza-CR²⁸. Sequential administration of DNMT and HDAC inhibitors resulted in clinical efficacy in patients with hematologic malignancies^{54,99}. However, other studies found no correlation between baseline methylation levels or methylation reversal and positive clinical outcome in patients with MDS or acute myeloid leukemia (AML) after combined treatment with 5-Aza-CR and entinostat¹⁰¹. The mechanism behind the clinical efficacy of sequential DNMT and HDAC inhibition remains controversial, and additional studies investigating potential genetic or epigenetic determinants of responsiveness will be helpful. Besides inducing apoptosis in cancer cells, another therapeutic approach involves inducing differentiation of cancer cells. To this end, following 5-Aza-CR and VPA treatment with all-*trans*-retinoic acid resulted in global hypomethylation and histone acetylation and clinical response in nearly half of treated patients with AML or high-risk MDS¹⁰².

Although targeting of histone demethylases is still in its infancy, early preclinical studies show promise for using such drugs alone (as described above) or together with other epigenetic therapies. Restoration of the expression of *SFRP2*, a negative regulator of Wnt signaling, in a human colon cancer model after LSD1 and DNMT inhibition has been associated with significant growth inhibition of established tumors⁶⁸. Notably, in addition to demethylating histone residues, LSD1 can demethylate DNMT1. This provides the ability to target both histone methylation and DNA methylation using a single compound. Cotreatment with the HDAC inhibitor panobinostat further enhances DZNep-mediated reduction in EZH2 levels, leading to increased p16, p21, p27 and FBX032 expression and apoptosis in cultured AML cells and mouse models¹⁰³. These promising data suggest that an absence of clinical trials targeting histone methylation and demethylation enzymes should not diminish enthusiasm for their therapeutic potential.

Epigenetic and cytotoxic therapies

Conventional chemotherapy can rapidly induce cell death in cancer cells, although resistance to standard chemotherapy often arises through epigenetic and DNA repair mechanisms²⁷. As a result, epigenetic therapeutics can be combined with more conventional therapies to induce responsiveness or overcome resistance to cytotoxic treatments. Preconditioning with epigenetic drugs could reverse the epigenetic alteration(s) that confer resistance, restoring chemotherapeutic sensitivity. For example, 5-Aza-CR treatment can reverse DNA methylation, thereby overcoming the gene silencing that led to chemotherapeutic resistance¹⁰⁴. In contrast, methylation-induced silencing of DNA repair genes, such as *MGMT*, is correlated with a positive clinical response to chemotherapy. Thus, the potential for success of combinations of DNA methylation inhibitors and chemotherapy may depend on the epigenetic profile of an individual tumor. Responses of patients with previously untreated non-small-cell lung cancer to combinations of the HDAC inhibitor vorinostat with carboplatin and paclitaxel were sufficiently promising to warrant a phase 2 study, which also showed encouraging results (Table 2)^{105,106}.

Conclusions

Several molecular regulators of the cellular epigenetic landscape have been established as effective targets in successful therapies for a variety of malignancies. In particular, the inhibition of DNMTs or HDACs has been approved for cancer treatment. Although the mechanism(s) behind the therapeutic benefit of DNMT and HDAC inhibition are not fully understood, ongoing and future studies that combine genomic sequencing and expression data may provide the keys to understanding the mechanism(s) underlying responsiveness. Besides their methylation and acetylation, histones can be phosphorylated, ubiquitinated and sumoylated. These modifications, which have been less well studied in the context of disease, may expand current possibilities for therapeutic intervention.

Given the importance of epigenetic mechanisms in controlling development and normal cellular behavior, it seems that approaches capable of targeting specific epigenetic alterations, rather than affecting global modifications, would greatly enhance clinical efficiency while lowering toxicity and side effects. This is an important priority for the field. Another major challenge in advancing epigenetic therapy will be to discriminate between so-called driver genes (those that must be epigenetically silenced for disease to occur) and so-called passenger genes (those that are epigenetically silenced owing to aberrant activity of the epigenetic machinery, but are not necessary for disease to occur). Recent advances in high-throughput technologies such as genome-wide sequencing, combined with RNA profiling, chromatin immunoprecipitation or bisulfite conversion, have generated large amounts of data that can be integrated to form a comprehensive understanding of the epigenetic alterations that are common and specific to various disease states. Assimilating these large datasets is likely to assist in identifying epigenetic alterations that are causative and those that are merely correlative^{107,108}. Thus, it may eventually be possible for patients to be screened, using high-throughput technologies, and classified by epigenetic alterations of the driver genes responsible for their illness. Along with the development of targeted inhibitors of epigenetic modifications, this could open the way for the use of personalized targeted therapies.

Currently, despite the successful clinical use of epigenetic therapies to treat hematological malignancies, there has been little success in treating solid cancers (Tables 2 and 3)^{109–112}. Initial clinical trials, which used treatment regimens later found to be less than optimal, resulted in low rates of positive clinical response. Administering more recently developed dosing and treatment schedules, and classifying tumor subtypes based on molecular signatures, may increase the efficacy of epigenetic therapy for solid tumors. Solid tumors invariably comprise heterogeneous populations of cells, many at different stages of differentiation. Clinical success may therefore require more effective approaches to determine which of these cells harbor epigenetic alterations and new strategies to ensure that therapeutic agents maintain stability and are able to penetrate the cellular mass and reach affected cells.

The recognition of epigenetics as a significant contributor to normal development and disease has opened new avenues for drug discovery and therapeutics, with a range of prospects that continues to expand as our knowledge of epigenetic regulation advances. Epigenetic therapies could be combined with conventional therapies to develop personalized treatments, render unresponsive tumors susceptible to treatment and reduce dosing. These advances may limit the side effects of treatment, improving compliance with dosing regimens and overall quality of life.

ACKNOWLEDGMENTS

Supported by R37CA082422 and R01CA083867 (P.A.J.). We thank members of the Jones laboratory for helpful discussions and careful reading of the manuscript, particularly H. Han for help in drawing chemical structures.

COMPETING FINANCIAL INTERESTS

The authors declare competing financial interests: details accompany the full-text HTML version of the paper at <http://www.nature.com/naturebiotechnology/>.

Published online at <http://www.nature.com/naturebiotechnology/>.

Reprints and permissions information is available online at <http://npg.nature.com/reprintsandpermissions/>.

- Portela, A. & Esteller, M. Epigenetic modifications and human disease. *Nat. Biotechnol.* **28**, 1057–1068 (2010).
- Meissner, A. Epigenetic modifications and their role in pluripotency. *Nat. Biotechnol.* **28**, 1079–1088 (2010).
- Lister, R. *et al.* Human DNA methylomes at base resolution show widespread epigenomic differences. *Nature* **462**, 315–322 (2009).
- Tahiliani, M. *et al.* Conversion of 5-methylcytosine to 5-hydroxymethylcytosine in mammalian DNA by MLL partner TET1. *Science* **324**, 930–935 (2009).
- Wu, S.C. & Zhang, Y. Active DNA demethylation: many roads lead to Rome. *Nat. Rev. Mol. Cell Biol.* **11**, 607–620 (2010).
- Jones, P.A. & Baylin, S.B. The epigenomics of cancer. *Cell* **128**, 683–692 (2007).
- Fouse, S.D. & Costello, J.F. Epigenetics of neurological cancers. *Future Oncol.* **5**, 1615–1629 (2009).
- Villeneuve, L.M. & Natarajan, R. The role of epigenetics in the pathology of diabetic complications. *Am. J. Physiol. Renal Physiol.* **299**, F14–F25 (2010).
- Javierre, B.M. *et al.* Changes in the pattern of DNA methylation associate with twin discordance in systemic lupus erythematosus. *Genome Res.* **20**, 170–179 (2010).
- Adcock, I.M., Ito, K. & Barnes, P.J. Histone deacetylation: an important mechanism in inflammatory lung diseases. *COPD* **2**, 445–455 (2005).
- Egger, G., Liang, G., Aparicio, A. & Jones, P.A. Epigenetics in human disease and prospects for epigenetic therapy. *Nature* **429**, 457–463 (2004).
- Urdinguio, R.G., Sanchez-Mut, J.V. & Esteller, M. Epigenetic mechanisms in neurological diseases: genes, syndromes, and therapies. *Lancet Neurol.* **8**, 1056–1072 (2009).
- Feng, J. & Fan, G. The role of DNA methylation in the central nervous system and neuropsychiatric disorders. *Int. Rev. Neurobiol.* **89**, 67–84 (2009).
- Sharma, S., Kelly, T.K. & Jones, P.A. Epigenetics in cancer. *Carcinogenesis* **31**, 27–36 (2010).
- Widschwendter, M. *et al.* Epigenetic stem cell signature in cancer. *Nat. Genet.* **39**, 157–158 (2007).
- Schlesinger, Y. *et al.* Polycomb-mediated methylation on Lys27 of histone H3 pre-marks genes for de novo methylation in cancer. *Nat. Genet.* **39**, 232–236 (2007).
- Ohm, J.E. *et al.* A stem cell-like chromatin pattern may predispose tumor suppressor genes to DNA hypermethylation and heritable silencing. *Nat. Genet.* **39**, 237–242 (2007).
- Shen, L. *et al.* Integrated genetic and epigenetic analysis identifies three different subclasses of colon cancer. *Proc. Natl. Acad. Sci. USA* **104**, 18654–18659 (2007).
- Seligson, D.B. *et al.* Global histone modification patterns predict risk of prostate cancer recurrence. *Nature* **435**, 1262–1266 (2005).
- Figuerola, M.E. *et al.* DNA methylation signatures identify biologically distinct subtypes in acute myeloid leukemia. *Cancer Cell* **17**, 13–27 (2010).
- Ellinger, J. *et al.* Global levels of histone modifications predict prostate cancer recurrence. *Prostate* **70**, 61–69 (2010).
- Bachmann, I.M. *et al.* EZH2 expression is associated with high proliferation rate and aggressive tumor subgroups in cutaneous melanoma and cancers of the endometrium, prostate, and breast. *J. Clin. Oncol.* **24**, 268–273 (2006).
- Weller, M. *et al.* MGMT promoter methylation in malignant gliomas: ready for personalized medicine? *Nat. Rev. Neurol.* **6**, 39–51 (2010).
- Kanai, Y. Genome-wide DNA methylation profiles in precancerous conditions and cancers. *Cancer Sci.* **101**, 36–45 (2010).
- Kondo, T. *et al.* Accumulation of aberrant CpG hypermethylation by *Helicobacter pylori* infection promotes development and progression of gastric MALT lymphoma. *Int. J. Oncol.* **35**, 547–557 (2009).
- Shen, L. *et al.* Drug sensitivity prediction by CpG island methylation profile in the NCI-60 cancer cell line panel. *Cancer Res.* **67**, 11335–11343 (2007).
- Ibanez de Caceres, I. *et al.* IGFBP-3 hypermethylation-derived deficiency mediates cisplatin resistance in non-small-cell lung cancer. *Oncogene* **29**, 1681–1690 (2010).
- Voso, M.T. *et al.* Valproic acid at therapeutic plasma levels may increase 5-azacytidine efficacy in higher risk myelodysplastic syndromes. *Clin. Cancer Res.* **15**, 5002–5007 (2009).
- Martens, J.W., Margossian, A.L., Schmitt, M., Foekens, J. & Harbeck, N. DNA methylation as a biomarker in breast cancer. *Future Oncol.* **5**, 1245–1256 (2009).

30. Jarmalaite, S. *et al.* Promoter hypermethylation in tumour suppressor genes and response to interleukin-2 treatment in bladder cancer: a pilot study. *J. Cancer Res. Clin. Oncol.* **136**, 847–854 (2010).
31. Baylin, S.B. & Ohm, J.E. Epigenetic gene silencing in cancer—a mechanism for early oncogenic pathway addiction? *Nat. Rev. Cancer* **6**, 107–116 (2006).
32. Comoglio, P.M., Giordano, S. & Trusolino, L. Drug development of MET inhibitors: targeting oncogene addiction and expedience. *Nat. Rev. Drug Discov.* **7**, 504–516 (2008).
33. Cheng, J.C. *et al.* Preferential response of cancer cells to zebularine. *Cancer Cell* **6**, 151–158 (2004).
34. Irizarry, R.A. *et al.* The human colon cancer methylome shows similar hypo- and hypermethylation at conserved tissue-specific CpG island shores. *Nat. Genet.* **41**, 178–186 (2009).
35. Issa, J.P. & Kantarjian, H.M. Targeting DNA methylation. *Clin. Cancer Res.* **15**, 3938–3946 (2009).
36. Fenaux, P. *et al.* Efficacy of azacitidine compared with that of conventional care regimens in the treatment of higher-risk myelodysplastic syndromes: a randomised, open-label, phase III study. *Lancet Oncol.* **10**, 223–232 (2009).
37. Yoo, C.B., Cheng, J.C. & Jones, P.A. Zebularine: a new drug for epigenetic therapy. *Biochem. Soc. Trans.* **32**, 910–912 (2004).
38. Yoo, C.B. *et al.* Delivery of 5-aza-2'-deoxycytidine to cells using oligodeoxynucleotides. *Cancer Res.* **67**, 6400–6408 (2007).
39. Toyota, M., Ohe-Toyota, M., Ahuja, N. & Issa, J.P. Distinct genetic profiles in colorectal tumors with or without the CpG island methylator phenotype. *Proc. Natl. Acad. Sci. USA* **97**, 710–715 (2000).
40. Toyota, M. *et al.* CpG island methylator phenotype in colorectal cancer. *Proc. Natl. Acad. Sci. USA* **96**, 8681–8686 (1999).
41. Tanemura, A. *et al.* CpG island methylator phenotype predicts progression of malignant melanoma. *Clin. Cancer Res.* **15**, 1801–1807 (2009).
42. Noshmehr, H. *et al.* Identification of a CpG island methylator phenotype that defines a distinct subgroup of glioma. *Cancer Cell* **17**, 510–522 (2010).
43. Van Rijnsoever, M., Elsaleh, H., Joseph, D., McCaul, K. & Iacopetta, B. CpG island methylator phenotype is an independent predictor of survival benefit from 5-fluorouracil in stage III colorectal cancer. *Clin. Cancer Res.* **9**, 2898–2903 (2003).
44. Soengas, M.S. *et al.* Inactivation of the apoptosis effector Apaf-1 in malignant melanoma. *Nature* **409**, 207–211 (2001).
45. Strathdee, G., MacKean, M.J., Illand, M. & Brown, R. A role for methylation of the hMLH1 promoter in loss of hMLH1 expression and drug resistance in ovarian cancer. *Oncogene* **18**, 2335–2341 (1999).
46. Ma, X., Ezzeldin, H.H. & Diasio, R.B. Histone deacetylase inhibitors: current status and overview of recent clinical trials. *Drugs* **69**, 1911–1934 (2009).
47. Fukushima, T., Takeshima, H. & Kataoka, H. Anti-glioma therapy with temozolomide and status of the DNA-repair gene MGMT. *Anticancer Res.* **29**, 4845–4854 (2009).
48. Wargo, J.A. *et al.* Recognition of NY-ESO-1+ tumor cells by engineered lymphocytes is enhanced by improved vector design and epigenetic modulation of tumor antigen expression. *Cancer Immunol. Immunother.* **58**, 383–394 (2009).
49. Wolff, E.M. *et al.* Hypomethylation of a LINE-1 promoter activates an alternate transcript of the MET oncogene in bladders with cancer. *PLoS Genet.* **6**, e1000917 (2010).
50. Feng, J. & Fan, G. The role of DNA methylation in the central nervous system and neuropsychiatric disorders. *Int. Rev. Neurobiol.* **89**, 67–84 (2009).
51. Shi, Y. Histone lysine demethylases: emerging roles in development, physiology and disease. *Nat. Rev. Genet.* **8**, 829–833 (2007).
52. Yoo, C.B. & Jones, P.A. Epigenetic therapy of cancer: past, present and future. *Nat. Rev. Drug Discov.* **5**, 37–50 (2006).
53. Nakagawa, M. *et al.* Expression profile of class I histone deacetylases in human cancer tissues. *Oncol. Rep.* **18**, 769–774 (2007).
54. Lane, A.A. & Chabner, B.A. Histone deacetylase inhibitors in cancer therapy. *J. Clin. Oncol.* **27**, 5459–5468 (2009).
55. Choudhary, C. *et al.* Lysine acetylation targets protein complexes and co-regulates major cellular functions. *Science* **325**, 834–840 (2009).
56. Haggarty, S.J., Koeller, K.M., Wong, J.C., Grozinger, C.M. & Schreiber, S.L. Domain-selective small-molecule inhibitor of histone deacetylase 6 (HDAC6)-mediated tubulin deacetylation. *Proc. Natl. Acad. Sci. USA* **100**, 4389–4394 (2003).
57. Balasubramanian, S. *et al.* A novel histone deacetylase 8 (HDAC8)-specific inhibitor PCI-34051 induces apoptosis in T-cell lymphomas. *Leukemia* **22**, 1026–1034 (2008).
58. Bruserud, Ø., Stapnes, C., Ersvaer, E., Gjertsen, B.T. & Rynningen, A. Histone deacetylase inhibitors in cancer treatment: a review of the clinical toxicity and the modulation of gene expression in cancer cell. *Curr. Pharm. Biotechnol.* **8**, 388–400 (2007).
59. Fischer, A. *et al.* Recovery of learning and memory is associated with chromatin remodeling. *Nature* **447**, 178–182 (2007).
60. Guan, J.S. *et al.* HDAC2 negatively regulates memory formation and synaptic plasticity. *Nature* **459**, 55–60 (2007).
61. Ptak, C. & Petronis, A. Epigenetics and complex disease: from etiology to new therapeutics. *Annu. Rev. Pharmacol. Toxicol.* **48**, 257–276 (2008).
62. Siu, L.L. Phase I study of MGC0103 given as a three-times-per-week oral dose in patients with advanced solid tumors. *J. Clin. Oncol.* **26**, 1940–1947 (2008).
63. Garcia-Manero, G. *et al.* Phase 1 study of the oral isotype specific histone deacetylase inhibitor MGC0103 in leukemia. *Blood* **112**, 981–989 (2008).
64. Lall, S. Primers on chromatin. *Nat. Struct. Mol. Biol.* **14**, 1110–1115 (2007).
65. Huang, J. & Berger, S.L. The emerging field of dynamic lysine methylation of non-histone proteins. *Curr. Opin. Genet. Dev.* **18**, 152–158 (2008).
66. Lee, Y.H. & Stallcup, M.R. Minireview: protein arginine methylation of nonhistone proteins in transcriptional regulation. *Mol. Endocrinol.* **23**, 425–433 (2009).
67. Metzger, E. *et al.* LSD1 demethylates repressive histone marks to promote androgen-receptor-dependent transcription. *Nature* **437**, 436–439 (2005).
68. Huang, Y. *et al.* Novel oligoamine analogues inhibit lysine-specific demethylase 1 and induce reexpression of epigenetically silenced genes. *Clin. Cancer Res.* **15**, 7217–7228 (2009).
69. Schulte, J.H. *et al.* Lysine-specific demethylase 1 is strongly expressed in poorly differentiated neuroblastoma: implications for therapy. *Cancer Res.* **69**, 2065–2071 (2009).
70. Wang, J. *et al.* The lysine demethylase LSD1 (KDM1) is required for maintenance of global DNA methylation. *Nat. Genet.* **41**, 125–129 (2009).
71. Gal-Yam, E.N. *et al.* Frequent switching of Polycomb repressive marks and DNA hypermethylation in the PC3 prostate cancer cell line. *Proc. Natl. Acad. Sci. USA* **105**, 12979–12984 (2008).
72. Mohammad, H.P. *et al.* Polycomb CBX7 promotes initiation of heritable repression of genes frequently silenced with cancer-specific DNA hypermethylation. *Cancer Res.* **69**, 6322–6330 (2009).
73. Tan, J. *et al.* Pharmacologic disruption of Polycomb-repressive complex 2-mediated gene repression selectively induces apoptosis in cancer cells. *Genes Dev.* **21**, 1050–1063 (2007).
74. Varambally, S. *et al.* Genomic loss of microRNA-101 leads to overexpression of histone methyltransferase EZH2 in cancer. *Science* **322**, 1695–1699 (2008).
75. Miranda, T.B. *et al.* DNep is a global histone methylation inhibitor that reactivates developmental genes not silenced by DNA methylation. *Mol. Cancer Ther.* **8**, 1579–1588 (2009).
76. Cha, T.L. *et al.* Akt-mediated phosphorylation of EZH2 suppresses methylation of lysine 27 in histone H3. *Science* **310**, 306–310 (2005).
77. Yu, J. *et al.* An integrated network of androgen receptor, polycomb, and TMPRSS2-ERG gene fusions in prostate cancer progression. *Cancer Cell* **17**, 443–454 (2010).
78. Huang, J. *et al.* G9a and Glp methylate lysine 373 in the tumor suppressor p53. *J. Biol. Chem.* **285**, 9636–9641 (2010).
79. Kondo, Y. *et al.* Downregulation of histone H3 lysine 9 methyltransferase G9a induces centrosome disruption and chromosome instability in cancer cells. *PLoS ONE* **3**, e2037 (2008).
80. Kubicek, S. *et al.* Reversal of H3K9me2 by a small-molecule inhibitor for the G9a histone methyltransferase. *Mol. Cell* **25**, 473–481 (2007).
81. Chang, Y. *et al.* Structural basis for G9a-like protein lysine methyltransferase inhibition by BIX-01294. *Nat. Struct. Mol. Biol.* **16**, 312–317 (2009).
82. Marango, J. *et al.* The MMSET protein is a histone methyltransferase with characteristics of a transcriptional corepressor. *Blood* **111**, 3145–3154 (2008).
83. Kim, H. *et al.* Requirement of histone methyltransferase SMYD3 for estrogen receptor-mediated transcription. *J. Biol. Chem.* **284**, 19867–19877 (2009).
84. Cloos, P.A. *et al.* The putative oncogene GASC1 demethylates tri- and dimethylated lysine 9 on histone H3. *Nature* **442**, 307–311 (2006).
85. Croce, C.M. Causes and consequences of microRNA dysregulation in cancer. *Nat. Rev. Genet.* **10**, 704–714 (2009).
86. Eacker, S.M., Dawson, T.M. & Dawson, V.L. Understanding microRNAs in neurodegeneration. *Nat. Rev. Neurosci.* **10**, 837–841 (2009).
87. Friedman, J.M. *et al.* The putative tumor suppressor microRNA-101 modulates the cancer epigenome by repressing the polycomb group protein EZH2. *Cancer Res.* **69**, 2623–2629 (2009).
88. Ng, E.K. *et al.* MicroRNA-143 targets DNA methyltransferases 3A in colorectal cancer. *Br. J. Cancer* **101**, 699–706 (2009).
89. Fabbri, M. *et al.* MicroRNA-29 family reverts aberrant methylation in lung cancer by targeting DNA methyltransferases 3A and 3B. *Proc. Natl. Acad. Sci. USA* **104**, 15805–15810 (2007).
90. Saito, Y. *et al.* Specific activation of microRNA-127 with downregulation of the proto-oncogene BCL6 by chromatin-modifying drugs in human cancer cells. *Cancer Cell* **9**, 435–443 (2006).
91. Lujambio, A. *et al.* A microRNA DNA methylation signature for human cancer metastasis. *Proc. Natl. Acad. Sci. USA* **105**, 13556–13561 (2008).
92. Kota, J. *et al.* Therapeutic microRNA delivery suppresses tumorigenesis in a murine liver cancer model. *Cell* **137**, 1005–1017 (2009).
93. McCaffrey, A.P. *et al.* The host response to adenovirus, helper-dependent adenovirus, and adeno-associated virus in mouse liver. *Mol. Ther.* **16**, 931–941 (2008).
94. Lanford, R.E. *et al.* Therapeutic silencing of microRNA-122 in primates with chronic hepatitis C virus infection. *Science* **327**, 198–201 (2010).
95. Yanai, N. *et al.* Unique microRNA molecular profiles in lung cancer diagnosis and prognosis. *Cancer Cell* **9**, 189–198 (2006).
96. Calin, G.A. & Croce, C.M. MicroRNA signatures in human cancers. *Nat. Rev. Cancer* **6**, 857–866 (2006).
97. Jahangeer, S., Elliott, R.M. & Henneberry, R.C. beta-Adrenergic receptor induction in HeLa cells: synergistic effect of 5-azacytidine and butyrate. *Biochem. Biophys. Res. Commun.* **108**, 1434–1440 (1982).

98. Cameron, E.E., Bachman, K.E., Myohanen, S., Herman, J.G. & Baylin, S.B. Synergy of demethylation and histone deacetylase inhibition in the re-expression of genes silenced in cancer. *Nat. Genet.* **21**, 103–107 (1999).
99. Kuendgen, A. & Lubbert, M. Current status of epigenetic treatment in myelodysplastic syndromes. *Ann. Hematol.* **87**, 601–611 (2008).
100. Chen, J., Odenike, O. & Rowley, J.D. Leukaemogenesis: more than mutant genes. *Nat. Rev. Cancer* **10**, 23–36 (2010).
101. Fandy, T.E. *et al.* Early epigenetic changes and DNA damage do not predict clinical response in an overlapping schedule of 5-azacytidine and entinostat in patients with myeloid malignancies. *Blood* **114**, 2764–2773 (2009).
102. Soriano, A.O. *et al.* Safety and clinical activity of the combination of 5-azacytidine, valproic acid, and all-trans retinoic acid in acute myeloid leukemia and myelodysplastic syndrome. *Blood* **110**, 2302–2308 (2007).
103. Fiskus, W. *et al.* Combined epigenetic therapy with the histone methyltransferase EZH2 inhibitor 3-deazaneplanocin A and the histone deacetylase inhibitor panobinostat against human AML cells. *Blood* **114**, 2733–2743 (2009).
104. Crea, F. *et al.* Epigenetic mechanisms of irinotecan sensitivity in colorectal cancer cell lines. *Mol. Cancer Ther.* **8**, 1964–1973 (2009).
105. Ramalingam, S.S. *et al.* Phase I and pharmacokinetic study of vorinostat, a histone deacetylase inhibitor, in combination with carboplatin and paclitaxel for advanced solid malignancies. *Clin. Cancer Res.* **13**, 3605–3610 (2007).
106. Ramalingam, S.S. *et al.* Carboplatin and Paclitaxel in combination with either vorinostat or placebo for first-line therapy of advanced non-small-cell lung cancer. *J. Clin. Oncol.* **28**, 56–62 (2010).
107. Satterlee, J., Schübeler, D. & Ng, H. Tackling the epigenome: challenges and opportunities for collaborative efforts. *Nat. Biotechnol.* **28**, 1039–1044 (2010).
108. Bernstein, B.E. *et al.* The NIH Roadmap epigenomics mapping consortium. *Nat. Biotechnol.* **28**, 1045–1048 (2010).
109. Rasheed, W., Bishton, M., Johnstone, R.W. & Prince, H.M. Histone deacetylase inhibitors in lymphoma and solid malignancies. *Expert Rev. Anticancer Ther.* **8**, 413–432 (2008).
110. Braithe, F. *et al.* Phase I study of epigenetic modulation with 5-azacytidine and valproic acid in patients with advanced cancers. *Clin. Cancer Res.* **14**, 6296–6301 (2008).
111. Balch, C., Fang, F., Matei, D.E., Huang, T.H. & Nephew, K.P. Minireview: epigenetic changes in ovarian cancer. *Endocrinology* **150**, 4003–4011 (2009).
112. Lin, J. *et al.* A phase I dose-finding study of 5-azacytidine in combination with sodium phenylbutyrate in patients with refractory solid tumors. *Clin. Cancer Res.* **15**, 6241–6249 (2009).
113. Berdasco, M. *et al.* Epigenetic inactivation of the Sotos overgrowth syndrome gene histone methyltransferase NSD1 in human neuroblastoma and glioma. *Proc. Natl. Acad. Sci. USA* **106**, 21830–21835 (2009).
114. Silverman, L.R. *et al.* Further analysis of trials with azacitidine in patients with myelodysplastic syndrome: studies 8421, 8921, and 9221 by the Cancer and Leukemia Group B. *J. Clin. Oncol.* **24**, 3895–3903 (2006).
115. Kantarjian, H.M. *et al.* Update of the decitabine experience in higher risk myelodysplastic syndrome and analysis of prognostic factors associated with outcome. *Cancer* **109**, 265–273 (2007).
116. Gore, S.D. *et al.* Impact of the putative differentiating agent sodium phenylbutyrate on myelodysplastic syndromes and acute myeloid leukemia. *Clin. Cancer Res.* **7**, 2330–2339 (2001).
117. Garcia-Manero, G. *et al.* Phase 1 study of the histone deacetylase inhibitor vorinostat (suberoylanilide hydroxamic acid [SAHA]) in patients with advanced leukemias and myelodysplastic syndromes. *Blood* **111**, 1060–1066 (2008).
118. Kelly, W.K. *et al.* Phase I study of an oral histone deacetylase inhibitor, suberoylanilide hydroxamic acid, in patients with advanced cancer. *J. Clin. Oncol.* **23**, 3923–3931 (2005).
119. Munster, P.N. *et al.* Phase I trial of vorinostat and doxorubicin in solid tumours: histone deacetylase 2 expression as a predictive marker. *Br. J. Cancer* **101**, 1044–1050 (2009).